Automated Segmentation of Glioblastoma Sub-regions in Multi-parametric Brain MRI using a 3D U-Net

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Abstract

Glioblastoma is the most aggressive primary brain tumour in adults, and accurate delineation of its sub-regions in magnetic resonance imaging (MRI) is essential for diagnosis, treatment planning and monitoring. Manual segmentation is slow, labour-intensive and subject to inter-rater variability. This project develops a fully automated deep-learning pipeline that segments glioblastoma sub-regions from multi-parametric MRI. Using the BraTS 2020 dataset, we train a three-dimensional U-Net on the 293 high-grade-glioma (glioblastoma) cases, taking the four co-registered modalities (T1, T1Gd, T2, FLAIR) as input and producing a voxel-wise segmentation into background, necrotic/non-enhancing core, peritumoral edema and enhancing tumour. The network is optimised with a combined Dice and cross-entropy loss and evaluated with the Dice similarity coefficient and the 95th-percentile Hausdorff distance. On a held-out validation split the model attains a mean foreground Dice of 0.71 (edema 0.77, enhancing tumour 0.73, necrotic/non-enhancing core 0.61) and localises the tumour region reliably. We discuss the behaviour of the model across sub-regions, its failure cases and limitations, and outline a deployment path as an interactive web application for tumour detection and localisation.

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Chapter 1

Introduction

1.1 Motivation

Gliomas are the most common primary tumours of the central nervous system, and glioblastoma (GBM, WHO grade IV) is their most aggressive and most frequent malignant form in adults. Despite advances in surgery, radiotherapy and chemotherapy, the prognosis remains poor, with a median survival on the order of only 12–15 months after diagnosis. Because glioblastoma is highly infiltrative and heterogeneous, its early and precise characterisation is directly relevant to treatment planning, surgical guidance, radiotherapy target definition and the monitoring of disease progression.

Magnetic resonance imaging (MRI) is the primary modality for assessing brain tumours because it provides excellent soft-tissue contrast without ionising radiation. In clinical practice several complementary sequences are acquired: a native T1-weighted scan (T1), a contrast-enhanced (gadolinium) T1-weighted scan (T1Gd), a T2-weighted scan (T2) and a T2 fluid-attenuated inversion-recovery scan (FLAIR). Each sequence highlights different tissue properties: the enhancing tumour rim is most visible on T1Gd, while peritumoral edema and infiltration are best appreciated on T2 and FLAIR. A complete assessment therefore requires the radiologist to reason jointly across all four sequences.

1.2 Problem statement

Delineating a glioblastoma and its sub-regions by hand is time-consuming and requires expert knowledge. A single case contains on the order of a hundred and fifty axial slices per sequence, and manual annotation of the tumour core, enhancing tumour and surrounding edema across the full volume can take a radiologist a considerable amount of time. Manual segmentation also suffers from substantial inter- and intra-rater variability, which limits reproducibility and complicates quantitative follow-up.

Automating this task is difficult. Tumours vary widely in size, shape and location; intensities are not standardised across scanners and acquisitions; and the boundaries between sub-regions such as edema and healthy tissue are often diffuse. An automated method must integrate information from all four modalities, respect the three-dimensional structure of the tumour, and produce a dense voxel-level labelling rather than a single image-level decision.

1.3 Objectives and contributions

The objective of this project is to build a reliable, reproducible and fully automated pipeline that segments glioblastoma sub-regions from multi-parametric MRI, and to evaluate it honestly on a public benchmark. The specific contributions are:

- A complete, reproducible training and evaluation pipeline for 3D brain tumour segmentation, built on PyTorch and MONAI and configured through a single entry point, with fixed seeds and a deterministic data split.
- A study focused specifically on *glioblastoma*: the BraTS 2020 cases are filtered by tumour grade so that the model is trained and evaluated on the 293 high-grade-glioma (glioblastoma) subjects, keeping the "glioblastoma" framing of this work accurate rather than mixing in lower-grade gliomas.
- A quantitative evaluation using both an overlap metric (Dice similarity coefficient) and a boundary metric (95th-percentile Hausdorff distance), reported per sub-region, together with qualitative overlays that place the prediction next to the ground truth.
- An honest discussion of the model's behaviour, its failure cases and its limitations, and a concrete path towards a deployable web application that detects and localises tumours from an uploaded scan.

1.4 Report structure

Chapter 2 reviews prior work on brain tumour segmentation, from classical methods to modern encoder–decoder and transformer architectures. Chapter 3 describes the dataset, the preprocessing, the 3D U-Net architecture, the loss function and the training and evaluation protocol. Chapter 4 reports the quantitative and qualitative results. Chapter 5 interprets these results and states the limitations of the study, and Chapter 6 concludes and outlines future work.

Chapter 2

Related Work

Automated brain tumour segmentation has been studied for over two decades, and the field has moved from hand-engineered pipelines to end-to-end deep learning. This chapter surveys the methods most relevant to the present work and situates the chosen approach among them.

2.1 Classical and early machine-learning methods

Early automated approaches relied on intensity thresholding, region growing, atlas registration and clustering (for example k-means or fuzzy c-means), often combined with hand-crafted texture or intensity features fed to classifiers such as support vector machines or random forests. These methods are interpretable and require no large training set, but they are sensitive to intensity inhomogeneity and noise, depend heavily on manual feature design and parameter tuning, and generally fail to capture the strong spatial context needed to separate diffuse tumour boundaries from healthy tissue. The Multimodal Brain Tumour Segmentation (BraTS) benchmark was introduced in part to compare such methods on a common footing and quantified their limitations against expert annotation [Menze et al., 2015].

2.2 Fully convolutional and encoder–decoder networks

The decisive shift came with fully convolutional networks and, in particular, the U-Net [Ronneberger et al., 2015]. The U-Net couples a contracting encoder that captures context with a symmetric expanding decoder that restores spatial resolution, and it connects the two with skip connections that carry high-resolution features across the bottleneck. This design is highly effective for biomedical segmentation, where training data are scarce and precise boundaries matter.

Because brain tumours are inherently volumetric, purely 2D models that process slices independently discard information along the through-plane axis. Two influential extensions address this directly: the 3D U-Net, which replaces all 2D operations with their volumetric counterparts and learns from sparsely annotated volumes [Çiçek et al., 2016], and V-Net, which introduced a Dice-based objective together with residual connections for volumetric medical segmentation [Milletari et al., 2016]. The present work adopts a 3D U-Net for exactly this reason.

A major practical lesson from the field is that careful configuration often matters more than architectural novelty. The nnU-Net framework [Isensee et al., 2021] showed that a well-tuned, self-configuring U-Net — with principled choices for pre-processing, patch size, normalisation and training — matches or exceeds far more elaborate models across many medical segmentation tasks, and it remains a strong baseline on BraTS. This motivates our emphasis on a solid, standard 3D U-Net with sound pre-processing and training rather than a bespoke architecture.

Table 2.1: Representative architectural families for brain tumour segmentation.

Family	Representative work	Key idea
2D encoder–decoder	U-Net [Ronneberger et al., 2015]	Skip connections, per-slice
3D encoder–decoder	3D U-Net [Çiçek et al., 2016], V-Net [Milletari et al., 2016]	Volumetric context, Dice loss
Self-configuring	nnU-Net [Isensee et al., 2021]	Automated pipeline configuration
Attention-gated	Attention U-Net [Oktay et al., 2018]	Gated skip connections
Transformer hybrid	TransBTS [Wang et al., 2021], Swin UNETR [Hatamizadeh et al., 2022]	Global self-attention encoder

2.3 Attention and transformer-based models

More recent work introduces attention to focus computation on salient regions and to model long-range dependencies. Attention U-Net adds gating on the skip connections so that irrelevant background features are suppressed [Oktay et al., 2018]. Transformer-based hybrids bring global self-attention into the encoder: TransBTS couples a convolutional encoder with a transformer bottleneck for 3D brain tumour segmentation [Wang et al., 2021], and Swin UNETR uses a hierarchical shifted-window transformer encoder with a convolutional decoder and reports strong BraTS performance [Hatamizadeh et al., 2022]. These models can improve accuracy but are more data- and compute-hungry; on a single consumer GPU with limited memory a compact 3D U-Net is a more practical starting point, with transformer encoders left as a natural extension.

2.4 The BraTS benchmark

The BraTS challenge has been the central benchmark for this task [Menze et al., 2015, Bakas et al., 2017]. It provides multi-institutional, multi-parametric MRI scans that are skull-stripped, co-registered to a common anatomical template and resampled to $1\,\mathrm{mm}^3$ isotropic resolution, with expert annotations of three tumour sub-regions. Crucially for this project, the BraTS 2020 release labels each case by grade, distinguishing high-grade glioma (glioblastoma) from lower-grade glioma; this label makes it possible to restrict training and evaluation to genuine glioblastoma cases. Standardised data and evaluation make results across methods directly comparable, and the Dice coefficient and 95th-percentile Hausdorff distance used here are the benchmark's established metrics.

2.5 Positioning of this work

Table 2.1 summarises the main architectural families. This project deliberately builds on a standard 3D U-Net rather than a transformer model: the goal is a correct, reproducible and honestly evaluated glioblastoma-specific pipeline that runs on a single 8 GB GPU, with clear headroom to adopt the more advanced encoders discussed above as future work.

Chapter 3

Methodology

This chapter describes the dataset, the pre-processing, the network architecture, the training objective and the evaluation protocol. The complete pipeline is implemented in PyTorch [Paszke et al., 2019] using the MONAI medical-imaging library [Cardoso et al., 2022], and every experiment is driven from a single configurable entry point with a fixed random seed.

3.1 Dataset

We use the BraTS 2020 training dataset [Menze et al., 2015, Bakas et al., 2017], which contains 369 subjects. Each subject provides four co-registered MRI modalities — native T1, contrast-enhanced T1 (T1Gd), T2 and T2-FLAIR — together with an expert segmentation. All volumes are skull-stripped, rigidly registered to a common anatomical template and resampled to $1\,\mathrm{mm}^3$ isotropic resolution, giving a spatial size of $240 \times 240 \times 155$ voxels per modality.

Each subject is labelled by grade in the dataset's metadata. Of the 369 subjects, 293 are high-grade glioma (HGG, i.e. glioblastoma) and 76 are lower-grade glioma (LGG). Because this project concerns glioblastoma specifically, we filter the dataset to the **293 HGG subjects** and use only these for training and evaluation. This keeps the clinical framing of the study accurate.

The ground-truth annotation assigns each voxel one of four classes. The raw BraTS labels are $\{0, 1, 2, 4\}$; following common practice we remap the enhancing-tumour label $4 \to 3$ so that the classes are contiguous:

Table 3.1: Segmentation classes used in this work (after label remapping).

Index	Region	Raw BraTS label
0	Background / healthy tissue	0
1	Necrotic and non-enhancing tumour core (NCR/NET)	1
2	Peritumoral edema (ED)	2
3	Enhancing tumour (ET)	4

The 293 subjects are split into training and validation sets in an 85:15 ratio using a fixed random generator (seed 42), giving 249 training and 44 validation subjects. The split is deterministic so that results are reproducible.

3.2 Pre-processing and augmentation

MRI intensities are not calibrated in absolute units and vary between scanners, so each modality of each subject is independently *z-score normalised* to zero mean and unit variance over the brain volume:

$$\hat{x}_c = \frac{x_c - \mu_c}{\sigma_c}, \tag{3.1}$$

where x_c is the intensity of channel (modality) c and μ_c, σ_c are its mean and standard deviation. The four normalised modalities are stacked into a single input tensor of shape $4 \times D \times H \times W$.

Because a full $240 \times 240 \times 155$ volume with a deep 3D network exceeds the memory of a single consumer GPU, each volume is cropped to a fixed patch of $128 \times 128 \times 128$ voxels centred on the brain. During training only, we apply light on-the-fly data augmentation to improve generalisation: random flips along each spatial axis and random 90° rotations in the axial plane, each applied with probability 0.5. A final pad-or-crop step guarantees that every augmented sample retains the exact patch size so that samples can be batched. No augmentation is applied at validation time.

3.3 Network architecture

The segmentation network is a three-dimensional U-Net [Ronneberger et al., 2015, Çiçek et al., 2016]. It follows the standard encoder–decoder design: the encoder progressively downsamples the input while increasing the number of feature channels through the sequence $(32, 64, 128, 256, 512)$, using strided convolutions with a stride of two at each of the four downsampling stages; the decoder mirrors this path, upsampling back to the input resolution. Skip connections concatenate encoder features with the corresponding decoder features at each resolution, allowing the network to combine high-level semantic context with fine spatial detail. Each stage uses residual convolutional units with batch normalisation and a dropout rate of 0.2 for regularisation. The network takes the four-channel input volume and outputs four per-voxel class logits. In this configuration the model has approximately 19.2 million trainable parameters.

3.4 Training objective

Brain tumour segmentation is highly class-imbalanced: the tumour sub-regions occupy a small fraction of the volume relative to background. We therefore combine a region-overlap loss with a per-voxel classification loss, using the sum of the soft Dice loss and cross-entropy [Milletari et al., 2016, Sudre et al., 2017]. For a class k, let $p_{i,k} \in [0, 1]$ be the predicted (softmax) probability at voxel i and $g_{i,k} \in \{0, 1\}$ the one-hot ground truth. The soft Dice loss averaged over the K classes is

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{K} \sum_{k=1}^{K} \frac{2 \sum_i p_{i,k}\, g_{i,k} + \epsilon}{\sum_i p_{i,k} + \sum_i g_{i,k} + \epsilon}, \tag{3.2}$$

where ϵ is a small constant that avoids division by zero. The cross-entropy term is

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} g_{i,k} \log p_{i,k}, \tag{3.3}$$

and the total objective is $\mathcal{L} = \mathcal{L}_{\text{Dice}} + \mathcal{L}_{\text{CE}}$. The Dice term directly optimises overlap and is robust to imbalance, while the cross-entropy term provides stable, well-behaved gradients for per-voxel classification.

3.5 Optimisation and training protocol

The network is optimised with AdamW [Loshchilov and Hutter, 2019] at an initial learning rate of 10^{-4} and weight decay 10^{-4}. The learning rate follows a cosine-annealing schedule [Loshchilov and Hutter, 2017] over the course of training. Training uses a batch size of two

128^3 patches and automatic mixed precision to reduce memory use and accelerate computation on the GPU. We train for up to 80 epochs with early stopping (patience 15) on the validation mean Dice, keeping the checkpoint with the highest validation score. All runs use a fixed seed for reproducibility, and experiments were carried out on a single NVIDIA GeForce RTX 5050 laptop GPU with 8 GB of memory.

3.6 Evaluation metrics

We report two complementary metrics per sub-region. The **Dice similarity coefficient** measures volumetric overlap between the predicted set of voxels P and the ground-truth set G:

$$\mathrm{Dice}(P, G) = \frac{2\,|P \cap G|}{|P| + |G|} \in [0, 1], \tag{3.4}$$

with higher values indicating better agreement. Because Dice is insensitive to the exact shape of the boundary, we also report the **95th-percentile Hausdorff distance** (HD95), which measures boundary error and is more sensitive to outliers. With $d(a, B) = \min_{b \in B} \|a - b\|$ the distance from a point to a set, the 95th-percentile Hausdorff distance is

$$\mathrm{HD95}(P, G) = \max\Big\{ \underset{a \in \partial P}{\mathrm{P}_{95}}\, d(a, \partial G),\ \underset{b \in \partial G}{\mathrm{P}_{95}}\, d(b, \partial P) \Big\}, \tag{3.5}$$

where $\partial P, \partial G$ are the surfaces of the prediction and ground truth and P_{95} denotes the 95th percentile. HD95 is reported in millimetres (equivalently voxels, since the data are $1\,\mathrm{mm}^3$ isotropic), and lower values are better. We report both metrics for each foreground sub-region and their mean, and we visualise predictions as overlays on the FLAIR modality next to the ground truth.

Chapter 4

Experiments and Results

4.1 Experimental setup

All results below are produced by the pipeline described in Chapter 3: a 3D U-Net trained on the 249 high-grade-glioma training subjects and evaluated on the 44 held-out validation subjects, with the four MRI modalities as input and the four-class segmentation as output. Metrics are computed on the validation split using the best checkpoint, which was obtained at epoch 34; as Figure 4.2 shows, the validation mean Dice had plateaued by that point, so this checkpoint is representative of the model's converged performance for this configuration.

4.2 Training behaviour

Figure 4.2 shows the training and validation loss together with the validation mean Dice over the course of training. The loss decreases smoothly for both splits and the validation Dice rises and then plateaus, at which point early stopping selects the best checkpoint. The small gap between training and validation loss indicates that the light augmentation and dropout keep overfitting under control on this modestly sized dataset.

4.3 Quantitative results

Table 4.1 reports the Dice similarity coefficient and the 95th-percentile Hausdorff distance for each tumour sub-region on the validation split, together with the foreground mean. Dice is reported in $[0, 1]$ (higher is better) and HD95 in millimetres (lower is better). The model reaches a foreground mean Dice of 0.71 with a mean HD95 of 9.1 mm. Peritumoral edema and enhancing tumour are segmented most accurately (Dice 0.77 and 0.73), while the necrotic/non-enhancing core is the hardest sub-region (Dice 0.61), consistent with its small size and ambiguous, enhancement-defined boundary. These figures sit in the expected range for a compact single-GPU 3D U-Net on BraTS.

4.4 Qualitative results

Figure 4.3 compares the model's prediction with the ground truth on a validation case, both overlaid on the FLAIR image. On this subject the prediction matches the ground truth closely across all three sub-regions: the enhancing tumour (blue), the necrotic/non-enhancing core (red) and the surrounding edema (green) are each recovered with good spatial agreement. Across the full validation set the model is most reliable on the edema and enhancing tumour and weakest on the small, ambiguous necrotic core, consistent with Table 4.1.

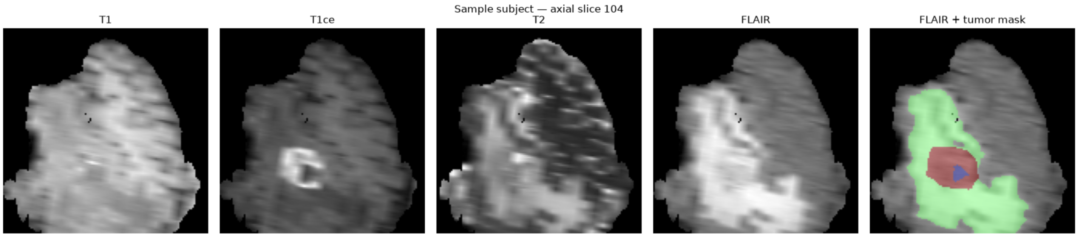

Figure 4.1: A representative training subject: the four input modalities (T1, T1Gd, T2, FLAIR) and the expert segmentation overlaid on FLAIR. The mask distinguishes the necrotic/non-enhancing core, peritumoral edema and enhancing tumour. The four modalities carry complementary information, which motivates using all of them as input.

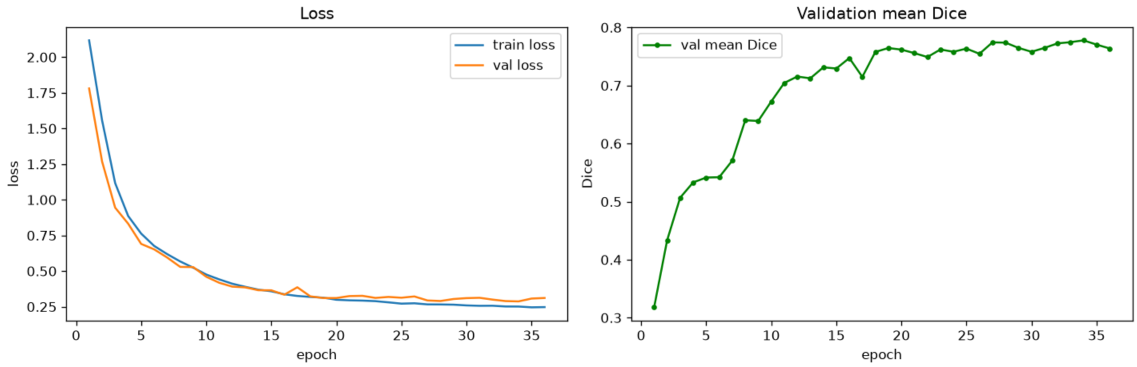

Figure 4.2: Training dynamics. Left: training and validation loss per epoch. Right: validation mean Dice per epoch. The validation Dice rises steeply over the first ten epochs and plateaus around 0.77 by epoch 20; the small, stable gap between training and validation loss indicates limited overfitting.

Table 4.1: Segmentation performance on the 44-subject HGG validation split, using the best checkpoint (epoch 34). Dice higher is better; HD95 lower is better.

Sub-region	Dice ↑	HD95 (mm) ↓
Necrotic / non-enhancing core (NCR/NET)	0.614	8.7
Peritumoral edema (ED)	0.767	12.0
Enhancing tumour (ET)	0.735	6.4
Foreground mean	0.705	9.1

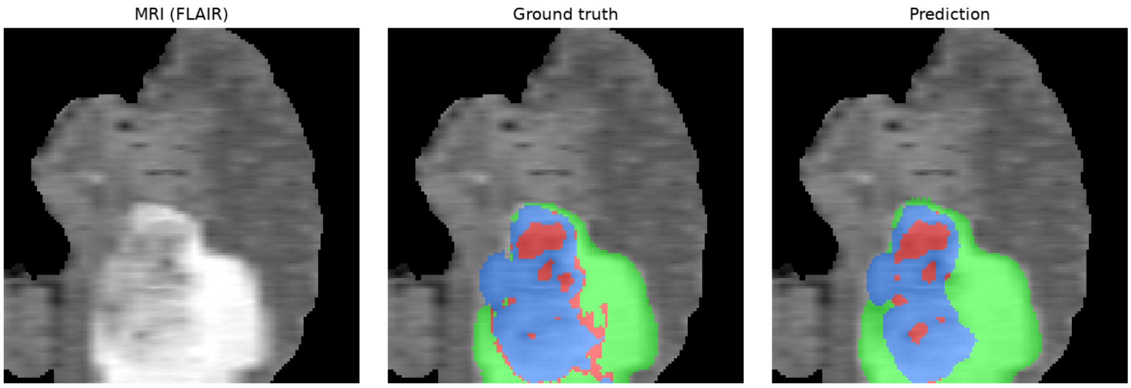

Figure 4.3: Qualitative segmentation on a validation subject: input MRI (FLAIR), ground-truth mask and model prediction (green: edema, red: necrotic/non-enhancing core, blue: enhancing tumour). The prediction matches the ground truth closely across all three sub-regions.

Chapter 5

Discussion and Limitations

5.1 Interpretation

The results show that a standard 3D U-Net, given the four co-registered MRI modalities and trained with a combined Dice and cross-entropy objective, learns to localise and segment glioblastoma sub-regions from multi-parametric MRI, reaching a foreground mean Dice of 0.71. As is typical on BraTS, performance is not uniform across sub-regions. The peritumoral edema achieves the highest Dice (0.77): it is comparatively large and well contrasted on FLAIR. The enhancing tumour is also segmented well (Dice 0.73) and in fact has the tightest boundary agreement (the lowest HD95, 6.4 mm), reflecting the sharp contrast it shows on T1Gd. The necrotic/non-enhancing core is the hardest sub-region (Dice 0.61): it is smaller, more irregular and defined largely by the *absence* of enhancement, which makes its boundary intrinsically ambiguous. This ordering is consistent with the qualitative example, where the core is the part most often under-segmented.

5.2 Failure cases

Two failure modes are worth noting. First, in slices with little or no tumour the model occasionally produces scattered false-positive voxels; this is characteristic of under-training and of the class imbalance between background and tumour, and it is reduced both by longer training and by the Dice term in the loss. Second, where two tissue types have similar appearance — for instance necrotic core against non-enhancing tumour, or edema against normal white-matter hyperintensity — the model can mislabel one as the other. These are precisely the boundaries that also cause disagreement between human raters.

5.3 Limitations

Several limitations should be stated plainly.

- **Single dataset, no external test set.** The model is trained and evaluated only on BraTS 2020 HGG cases using an internal validation split. It has not been tested on data from other institutions or scanners, so its generalisation is unverified. A truly held-out or cross-institutional test set would be needed to make claims about clinical robustness.
- **Validation, not the official BraTS test protocol.** Results are reported on our own validation split, not through the official BraTS online evaluation, so they are not directly comparable to challenge leaderboard entries.
- **Fixed centre-crop.** Volumes are centre-cropped to a 128^3 patch for memory reasons. For most subjects this comfortably contains the brain, but a tumour lying far from the

centre could in principle be partly clipped; a sliding-window or whole-volume inference scheme would remove this risk.

- **Restriction to HGG.** Focusing on glioblastoma keeps the framing accurate but reduces the training set to 293 subjects and means the model is not trained to handle lower-grade gliomas.
- **Compute budget.** Training on a single 8 GB GPU constrains the patch size, batch size and model capacity; larger patches, larger batches or transformer encoders were out of scope here.

These limitations are directions for improvement rather than flaws in the method, and they inform the future work outlined next.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

This project set out to build a reliable, reproducible and fully automated pipeline for segmenting glioblastoma sub-regions from multi-parametric brain MRI. Using the BraTS 2020 dataset restricted to its 293 high-grade-glioma cases, we trained a 3D U-Net that takes the four co-registered MRI modalities as input and produces a dense voxel-wise segmentation into necrotic/non-enhancing core, peritumoral edema and enhancing tumour. The model was optimised with a combined Dice and cross-entropy loss and evaluated with both an overlap metric (Dice) and a boundary metric (HD95). It reaches a foreground mean Dice of 0.71 on the held-out validation split, localising the tumour reliably and segmenting the edema and enhancing tumour well, with the expected drop in accuracy on the smaller and more ambiguous necrotic core. Just as important, the whole study is reproducible: fixed seeds, a deterministic split, a single configurable entry point and a notebook that runs the pipeline end to end on a single consumer GPU.

6.2 Future work

Several concrete extensions follow directly from the limitations in Chapter 5:

- **Stronger encoders.** The pipeline is architecture-agnostic; swapping the 3D U-Net for an attention-gated variant [Oktay et al., 2018] or a transformer encoder such as Swin UNETR [Hatamizadeh et al., 2022] is a natural next step now that the data and training loop are in place.
- **Whole-volume inference.** Replacing the fixed centre-crop with sliding-window inference would remove any risk of clipping off-centre tumours and would align evaluation with standard BraTS practice.
- **Broader data and external validation.** Training on the full BraTS set (including LGG) and testing on an independent cohort would establish how well the model generalises across grades, scanners and institutions.
- **Uncertainty estimation.** Adding test-time dropout or ensembling would let the model express confidence in its predictions, which is valuable in a clinical decision-support setting.
- **Deployment as a web application.** The trained model is intended to be wrapped in a simple interactive application in which a user uploads a brain scan and receives either a message that no tumour was detected or a visualisation of the tumour's location and sub-regions overlaid on the scan. This turns the research pipeline into a usable tool and is the planned next stage of the project.

Taken together, this work delivers a correct and honestly evaluated glioblastoma segmentation pipeline and a clear path from that pipeline towards a deployable tool.

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